

Computer Knowledge—Engineering Level: Operating-System Concepts

Subject: Computer Knowledge

An operating system is system software that manages computer hardware and provides common services to application programs. It creates an abstraction layer so that programmers can work with files, processes, memory, and devices without controlling every electrical detail directly. Common operating-system responsibilities include process management, memory management, file systems, device control, security, networking, and user interaction. The kernel is the privileged core that manages resources and responds to requests from applications.

A program is passive code stored on a device, while a process is a running instance of a program with its own execution state. A process has an address space, registers, a program counter, open resources, and other operating-system data. Threads are execution paths within a process. Threads in the same process commonly share memory, which can make communication efficient but can also create race conditions. A race condition occurs when the result depends on the timing of concurrent operations. Locks, atomic operations, message passing, and carefully designed data structures help coordinate access to shared state.

The operating system schedules processes and threads on available CPU cores. A scheduler chooses which runnable task should execute and for how long. Scheduling policies may attempt to balance responsiveness, fairness, throughput, and priority. A task can be running, ready, blocked while waiting for an event, or terminated. A context switch changes the CPU from one task to another. Context switching supports multitasking but consumes time, so excessive switching can reduce performance.

Virtual memory allows each process to use a logical address space that is mapped to physical memory. Pages can be moved between RAM and storage when necessary. This gives programs isolation and can allow them to use more address space than the available physical RAM, but excessive paging causes slow performance. Memory protection prevents one process from reading or overwriting another process's memory. The file system organizes persistent data into files and directories and records metadata such as names, permissions, locations, and timestamps.

Operating systems also manage devices through drivers and interfaces. A driver translates general operating-system requests into device-specific operations. Interrupts allow hardware to notify the CPU when an event needs attention, such as a completed disk operation or incoming network data. Security features include user accounts, permissions, privilege separation, secure boot, process isolation, and auditing. Engineering systems must consider failures: a process may crash, a disk may become unavailable, or a network request may time out. Good designs use logs, monitoring, backups, safe defaults, and recovery procedures. Understanding

operating-system concepts helps engineers build software that is reliable, efficient, concurrent, and secure.